Full Market Depth

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Level 3 data includes market depth upto 20 levels. We are extending beyond and adding **200 level data**. This shows complete picture of the market movements and it is streamed real-time via websockets.

This data can be used to detect demand supply zones, outside of 5 level market depth and build trading systems to detect market movements.

> Only NSE Equity and Derivatives segments are enabled for Full Market Depth.

Similar to Live Market Feed, all request messages over WebSocket are in JSON whereas all response messages over WebSocket are in Binary.

Establishing Connection

20 Level

To establish connection with DhanHQ WebSocket for 20 Level Market Depth, you can connect to the below endpoint using WebSocket library.

```
wss://depth-api-feed.dhan.co/twentydepth?token=<your-access-token>&clientId=<your-client-id>&authType=2
```

200 Level

To establish connection with DhanHQ WebSocket for 200 Level Market Depth, you can connect to the below endpoint using WebSocket library.

```
wss://full-depth-api.dhan.co/twohundreddepth?token=<your-access-token>&clientId=<your-client-id>&authType=2
```

Query Parameters

FIELD	DESCRIPTION
token*	Access Token generated via Dhan
clientId*	User specific identification generated by Dhan
authType*	2 by Default

Adding Instruments

20 Level

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For 20 Level Market Depth, you can subscribe upto 50 instruments in a single connection and receive market data packets.

For subscribing, this can be done using JSON message which needs to be sent over WebSocket connection.

> **Note**
>
> You can send all 50 instruments in a single JSON message for 20 Depth. You can send multiple messages over a single connection as well to subscribe to all instruments in parts and receive data.

Request Structure

```
{
  "RequestCode" : 23,
  "InstrumentCount" : 1,
  "InstrumentList" : [
    {
      "ExchangeSegment" : "NSE_EQ",
      "SecurityId" : "1333"
    }
  ]
}
```

Parameters

FIELD (REQUIRED*)	DESCRIPTION
RequestCode * int	Code for subscribing to particular data mode for Full Market Depth (see Annexure) 23
InstrumentCount * int	No. of instruments to subscribe from this request
InstrumentList.ExchangeSegment * enum string	Exchange Segment of instrument to be subscribed (see Annexure)
InstrumentList.SecurityId * string	Exchange standard ID for each scrip. refer here

200 Level

In 200 level market depth, only 1 instrument per connection can be subscribed. The JSON payload needs to be sent similar to 20 level depth subscription, while the socket connection has been established.

Request Structure

```
{
  "RequestCode" : 23,
```

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```
    "ExchangeSegment" : "NSE_EQ",
    "SecurityId" : "1333"
}
```

Parameters

FIELD (REQUIRED*)	DESCRIPTION
RequestCode * int	Code for subscribing to particular data mode for Full Market Depth (see Annexure) 23
ExchangeSegment * enum string	Exchange Segment of instrument to be subscribed (see Annexure)
SecurityId * string	Exchange standard ID for each scrip. refer here

Keeping Connection Alive

To keep the WebSocket connection alive and prevent it from closing, the server side uses **Ping-Pong** module. Server side sends ping every 10 seconds to the client server (in this case, your system) to maintain WebSocket status as open.

An automated pong is sent by websocket library. You can use the same as response to the ping.

> In case the client server does not respond for more than 40 seconds, the connection is closed from server side and you will have to reestablish connection.

Response Structure

The market depth data is sent as structured binary packet. It will require parsing to readable format to extract the relevant information.

All responses from Dhan Market Feed consists of Response Header and Payload. Header for every response message remains the same with different feed response code, while the payload can be different.

20 Level

Response Header

The response header message is of 12 bytes which will remain as part of the response message. The message structure is given as below.

BYTES	TYPE	SIZE	DESCRIPTION
0-1	int16	2	Message Length of the entire payload packet

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BYTES	TYPE	SIZE	DESCRIPTION
2	[ ] byte	1	Feed Response Code can be referred in Annexure
3	[ ] byte	1	Exchange Segment can be referred in Annexure
4-7	int32	4	Security ID - can be found here
8-11	uint32	4	Message Sequence (to be ignored)

Depth Packet Depth Data Packet for 20 level market depth is structured differently from 5 level depth. Over here, you will receive the bid (sell) and ask (buy) data packets separately, each containing 20 packets of 16 bytes each.

BYTES	TYPE	SIZE	DESCRIPTION
0-11	[ ] array	12	Response Header `41` for Bid Data (Buy) `51` for Ask Data (Sell) (see Annexure)
12-331	Bid/Ask Depth Structure	320	20 packets of 16 bytes each for each instrument in below provided structure

Each of these 20 packets will be received in the following packet structure:

BYTES	TYPE	SIZE	DESCRIPTION
0-7	float64	8	Price
8-11	uint32	4	Quantity
12-15	uint32	4	No. of Orders

200 Level

Response Header

The response header message is of 12 bytes which will remain as part of the response message. The message structure is given as below.

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BYTES	TYPE	SIZE	DESCRIPTION
0-1	int16	2	Message Length of the entire payload packet
2	[ ] byte	1	Feed Response Code can be referred in Annexure
3	[ ] byte	1	Exchange Segment can be referred in Annexure
4-7	int32	4	Security ID - can be found here
8-11	uint32	4	No of Rows - gives number of rows to be read for response

Depth Packet 200 level market depth is structured similar to 20 level depth. Over here, you will receive the bid (sell) and ask (buy) data packets separately, each containing multiple packets of 16 bytes each.

BYTES	TYPE	SIZE	DESCRIPTION
0-11	[ ] array	12	Response Header `41` for Bid Data (Buy) `51` for Ask Data (Sell) (see Annexure)
12-3211	Bid/Ask Depth Structure	3200	200 packets of 16 bytes each for each instrument in below provided structure

Each of these 200 packets will be received in the following packet structure:

BYTES	TYPE	SIZE	DESCRIPTION
0-7	float64	8	Price
8-11	uint32	4	Quantity
12-15	uint32	4	No. of Orders

Note

Whenever 20 or 200 level depth packets are sent on the connection, they are stacked one after another in a single message. For 20 level depth, if two instruments are subscribed, then the first instrument's Bid

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packet followed by Ask packet of that instrument is added and then the second instrument's bid and ask packets in same sequence. To handle this, you can break down the packet on the basis of length.

Feed Disconnect

Stopping a single instrument and closing the whole feed are two different operations, with two different request codes. They are not interchangeable.

Unsubscribe instruments — `25`

Request code `25` stops the depth feed for the instruments you name in the message. It must carry the **same request structure you used to subscribe**, including `InstrumentCount` and `InstrumentList`:

```
{
  "RequestCode" : 25,
  "InstrumentCount" : 1,
  "InstrumentList" : [
    {
      "ExchangeSegment" : "NSE_EQ",
      "SecurityId" : "11536"
    }
  ]
}
```

Connect the feed — `41`

Request code `41` opens the feed and nothing else. It carries no instrument list and subscribes to nothing — use it when you want the connection established on its own, then send subscribe messages separately.

```
{
  "RequestCode" : 41
}
```

Disconnect the feed — `11`

To close the entire feed, send request code `11` on its own. No instrument list is needed, and it ends the feed regardless of what you are subscribed to:

```
{
  "RequestCode" : 11
}
```

Server-side disconnection

In case of WebSocket disconnection from server side, you will receive disconnection packet, which will have disconnection reason code.

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- If more than 5 websockets are established, then the first socket will be disconnected with `805` with every additional connection.

BYTES	TYPE	SIZE	DESCRIPTION
0-11	[ ] array	12	Response Header with code `50` (see Annexure)
12-13	int16	2	Disconnection message code - here

You can find detailed Disconnection message code description here.

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